Jian Feng

j36feng@uwaterloo.ca | 647-766-0178 | Website | linkedin/in/fenjian | github/faketut

EMPLOYMENT

Manulife Financial Corporation

Co-op 2026 –

- Cut AKS namespace onboarding from days to minutes by automating ServiceNow tasks into Terraform/GitOps PRs with auto change tickets.
- Built a Foundry LLM agent that validates cert-SAN gitops PRs against runbooks across 9 AKS clusters, replacing manual review and catching misconfigs before prod.
- Built Power Apps-driven DL automation framework reducing manual membership updates to minutes via event-driven Graph API sync.

Treasury Board of Canada Secretariat

Co-op 2025

- Shipped a .NET MVC module auto-normalizing job titles across a 5K-firm public dataset, unblocking analyst peer-group and governance reporting.
- Migrated 30 Selenium cases (5 suites) to Katalon on Azure DevOps CI/CD; demoed to execs and adopted by other QA teams to ease no-code onboarding.
- Refactored core modules of the enterprise data platform and tuned SQL, cutting 5K-firm-facing browser page load 30% for faster reviews.

Lansi Institute of Artificial Intelligence in Medicine

Co-op (Remote) 2024

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Ran XGBoost + SMR/MR² across a 68-study T2D meta-analysis as research assistant; narrowed the candidate drug-target shortlist used in the published paper.

Sanyang Technology Co., Ltd.

Intern 2024

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

EDUCATION

University of Waterloo 2023-2028

Bachelor of Computing and Financial Management

- Majors: Computer Science, Finance, Statistics
- Grew UX Research Hub from 10→318 members as Director; secured Google UXR keynote at $0 club cost and TikTok chapter cross-promotion driving sustained inbound growth.
- Velocity Co-op Problem Award, 3rd in 12th SAF Investment Research Challenge, Finalist in 7th SAF Strategy Case Competition
- Certifications: Duke Data Engineering, IBM Vector Database, UCI Database Design & Operational BI, Google Data Analytics, SAS Time Series and Sequential Data Analysis, Stanford Machine Learning, UCI Mathematics for ML, UCSC Bayesian Statistics, UPenn Finance & Quantitative Modeling, Yonsei Valuation and Financial Analysis for Startups

PERSONAL PROJECTS

CabStream ETL

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

qmj-tsx

- Replicates Asness QMJ factor on TSX small-caps; price-derived Quality proxy plugs the sparse-fundamental gap via Parquet pipelines.
- Multi-source ingest (yfinance, AQR, Ken French) feeds Newey–West regressions and sector×weighting×subperiod robustness sweeps.

audited-pa

- IBKR Flex XML → monthly tear sheets and position-level attribution via pandas + pyarrow; qpdf redacts statements with a verification loop.
- Source-of-truth reconciliation re-derives every headline stat from raw fills, so audited cumulative history stays tamper-evident.

ArmyOfSafeguards

- Four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable via env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold baselines on held-out HF.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

TECHNOLGY

Python, SQL, C#, PostgreSQL, MySQL, MongoDB, AWS, GCP, Azure, Docker, Kubernetes, Power BI, Tableau, Excel,